Human Activity and Sign Gesture Recognition Using Wi-Fi Channel State Information for Inclusive Smart Homes

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***Abstract*—A smart home that accepts non-verbal commands has to tell deliberate signs apart from ordinary movement. We examine this problem with a single IEEE 802.11 link between two software-defined radios, using only the amplitude of the channel state information (CSI). Five participants in one room performed 12 daily activities and 14 signs taken from Indian Sign Language (ISL) references, both as isolated cued trials and in 34 scripted sequences in which a block of activities is followed by a block of signs. We evaluate a residual 1-D convolutional network with each participant held out of both training and model selection. When event boundaries and type are given, mean per-participant macro-F1 is 50.2% for isolated activities and 33.2% for isolated signs; a handcrafted-feature baseline reaches 22.1% on the same activity folds. In the scripted sequences, where the system has to find the signs by itself, it detects 121 of the 208 signs but names only 39 (18.8%) correctly, and it issues 50 unmatched commands during 20.3 minutes of routine activity. Most of the loss comes from separating signs from the activity around them, and seated movements are the weakest. These offline, preliminary results show how far known-boundary evaluation overstates performance in continuous use. The recordings, labels and evaluation folds are publicly released.**

***Index Terms*—Wi-Fi sensing, channel state information, activity recognition, sign gestures, continuous recognition, smart homes**

I. Introduction

Before a home can act on a sign, it has to answer a simpler question: was the movement meant as a message at all? Most movements in a home are not. Picking up a cup should not be read as a WATER request. Wi-Fi CSI suits this setting because it needs neither a camera nor a worn device. Body motion changes the multipath channel, and every decoded frame carries a channel estimate for each subcarrier [1].

Most Wi-Fi gesture and sign studies are evaluated on examples that have already been cut out of the recording. SignFi classifies 276 manually segmented sign gestures and leaves automatic segmentation to future work [2]. Its results also show how much the evaluation split matters. With the traces of five users mixed under 5-fold cross-validation it reports 86.66% on 150 signs, but leave-one-subject-out accuracy drops to 67.36%, and to 49.3% for one user recorded in a different setting [2]. High in-domain accuracy is common for activity recognition [3], while robustness across domains has so far relied on several synchronized receivers [4]. WiSee addressed accidental activation directly: it uses a repeated preamble gesture and reports false positives over 24 hours with the target user absent [5]. False commands raised while someone goes about ordinary activities near the link are a different and harder case.

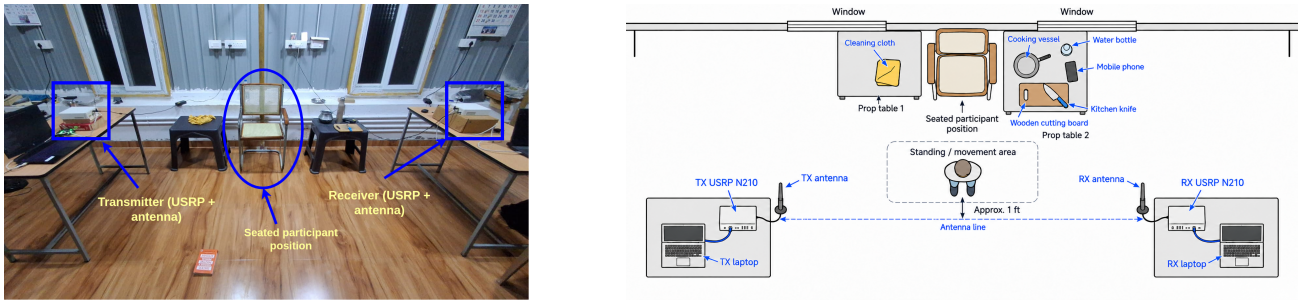
This work-in-progress paper asks a narrower question. On a single CSI link, how well can a model trained on other people separate daily activities from a chosen sign vocabulary, name them, and find and name signs in continuous recordings of a new person? We compare three levels of supplied information: isolated events with known boundaries and type, sequence steps with known boundaries and type, and fully automatic detection. The paper contributes (i) a documented five-person, single-link capture protocol that mixes activities and signs in both isolated and continuous recordings; (ii) a held-out-participant pipeline in which every model choice is made on a separate validation participant; (iii) a measurement, with error analysis, of the gap between known-boundary and automatic recognition; and (iv) the public release of the raw CSI, labels, documentation and the exact evaluation folds [8]. All results are offline, from one room, with one active person at a time.

II. Data Collection

A. Radio setup

Two Ettus USRP N210 radios with SBX daughterboards serve as transmitter and receiver. The receiver runs the GNU Radio IEEE 802.11 receiver [6], which we extended to log the least-squares channel estimate of the 52 occupied subcarriers for every decoded frame, stamped with the receiver's clock. The link uses channel 1 (2412 MHz) at 20 MS/s. The transmitter targets 200 frames/s, and the participant recordings gave about 90–131 decoded CSI rows/s. Because the two radios share no clock, only the amplitude is used. Transmit gain was 15 dB and receive gain 12 dB, except for P01's activity recording (16 dB). The receive gain was chosen from a sweep at fixed transmit gain, where the decoded rate peaked at 12 dB (21 rows/s at 0 dB, 128 at 12 dB, 117 at 16 dB, 57 at 24 dB). Antenna positions and the room stayed fixed, fans were off and nearby phones were in airplane mode.

Figure 1 shows the setup. Each USRP connects to a laptop; the chair and prop tables touch the back wall, and the standing reference is about 0.30 m behind the antenna line. The household items on the tables are activity props, not sensing inputs.



(a) Recording environment.

(b) Experimental layout.

Fig. 1. Single-link CSI collection: (a) recording environment; (b) equipment, props and participant positions. The dashed antenna line is a geometric reference, not a sensing boundary.

TABLE I
SCORED EVENTS BY RECORDING BLOCK

Block	People	Events	Labels
Isolated activities, standing/moving	5	488	10 or 12 classes
Isolated activities, seated	2	140	7 classes
Isolated signs, standing	5	350	14 classes
Isolated signs, seated	2	140	14 classes
Sequence activity steps (34 runs)	5	208	5–9 steps/run
Sequence sign steps (34 runs)	5	208	6–7 steps/run
Total	5	1,534	

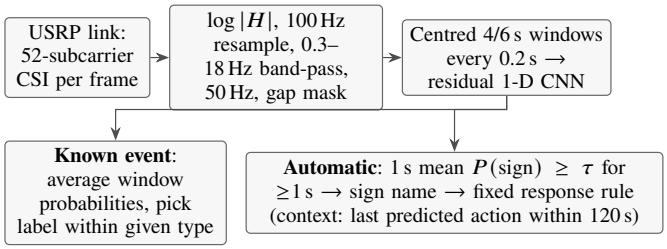


Fig. 2. Pipeline. The CNN has a 3-state head (idle/activity/sign) and a 26-label identity head. The causal filter does not make the system online: windows are centred, so each prediction uses 2–3 s of future signal.

B. Protocol and labels

Activities (12): walk, sit–stand, pick up, light switch, drink, answer phone, eat, wipe table, TV remote, fall (onto a mattress) and, for P04–P05 only, chop food and stir pot with cold props. *Signs (14):* HELLO, THANK-YOU, LAMP, FAN, TELEVISION, COOK, WATER, HOT, HUNGRY, TOILET, CALL, HELP, PAIN, ACCIDENT, chosen from INCLUDE [7], an ISL emergency-gesture dataset [9] and ISL dictionary entries. The linguistic validity of the performed signs has not been independently checked, and the vocabulary is a command set rather than continuous ISL.

In the isolated blocks, an on-screen cue marks each trial (4–6 s for activities, 4 s for signs), and the labels are the cue intervals on the receiver clock. P01–P03, recorded on one day, performed standing activity and sign blocks. P04–P05, recorded about two weeks later, also performed a seated activity block, a seated sign block and the two cooking activities. In the scripted sequences, an operator key press marks each step boundary. Three templates (meal, TV/phone, cleaning-then-fall) were recorded twice by every participant, and a fourth, fully seated meal template by P04–P05, giving 34 runs with 208 activity and 208 sign steps. Each template performs its activities first and its signs second, so the sequences are not arbitrarily interleaved. The labels have not been checked against video.

After excluding 27 P02 activity intervals that the recording does not cover, the evaluation contains 628 isolated activities, 490 isolated signs and 416 sequence steps from 22 participant recordings (2,010,899 CSI rows, 4.8 h including rest periods). Three empty-room recordings were used only for quality checks (Table I).

All five participants are authors of this paper and gave informed consent for the recordings and their public release. The recordings, labels, documentation and evaluation folds are on IEEE DataPort [8].

III. Recognition Pipeline

A. Preprocessing and windows

Per-subcarrier amplitude is log-transformed, linearly interpolated on the receiver timestamps to 100 Hz, passed through a causal second-order Butterworth band-pass filter (0.3–18 Hz) and decimated to 50 Hz (Fig. 2). Samples that bridge gaps longer than 100 ms are masked; a gap longer than 250 ms resets the filter and masks the following 3 s. Each subcarrier is scaled by its RMS over the training participants. Windows are centred every 0.2 s and kept only if at least 95% of the surrounding 6 s is valid. A window takes its state and identity labels from its central 2 s when at least 80% of those samples agree. Between isolated trials, gaps of up to 7 s are labelled *activity* in activity blocks, since participants walk back to their starting position, and *idle* in sign blocks; sequence steps carry only their own labels. No participant identifier, file name, true context or event duration is given to the model.

B. Model and training

All three candidates share a dilated residual 1-D CNN with four residual blocks (dilations 1–8) and global mean and max pooling. Two use batch normalization, with 4 s and

6 s windows. The third uses 4 s windows divided by their RMS energy, group normalization, and log energy and log temporal-difference energy appended to the pooled features. Each network predicts the 3-state output and the 26-label identity output, and identity is decided within actions or within signs. Training uses AdamW (learning rate 10^{-3}, weight decay 0.01, batch 256, cosine schedule with a 100-epoch horizon), label smoothing and gradient clipping, and samples with replacement so that state, class and participant are balanced. The augmentation applies a window gain of 0.65–1.5, a per-subcarrier gain of 0.85–1.15, random sign inversion, noise of up to 4% of the window RMS and 5% subcarrier dropout. For isolated activities only, the CNN probabilities may be averaged with an ExtraTrees classifier on handcrafted energy, band-power and envelope features.

In continuous recordings, the sign probability is averaged over the preceding 1 s of window positions. A run at or above a threshold τ that lasts at least 1 s becomes a command, and a run is split wherever a different sign identity persists for 1 s. Each command takes the sign with the highest mean probability. A fixed rule then maps it to a response, using as context the most recent predicted anchor action within 120 s; for example, HOT after eating maps to a kitchen-hazard alert and HELP after a fall to an emergency alert. The mapping is predefined; intent is not learned.

C. Evaluation protocol

Each of the five participants is tested once. Within each fold, a fixed second participant (test/validation: P01/P04, P02/P05, P03/P01, P04/P02, P05/P03) is used to choose, separately for the detector, activity-identity and sign-identity roles, the candidate network and its epoch count (early stopping, with validation every 4 epochs); the threshold $\tau \in [0.2, 0.9]$ that maximizes sequence-sign identity F1; and the ExtraTrees weight, from $\{0, 0.5, 1\}$. The chosen networks are then refitted on all four non-test participants with two seeds and averaged. The selections differed between folds. The energy-normalized network filled 10 of the 15 fold-role slots, τ ranged from 0.30 to 0.85, and only the P05 fold chose ExtraTrees fusion (weight 0.5). This is a fixed-validation design rather than exhaustive nested cross-validation, and the protocol was revised after an earlier run on the same data, so the folds are not an untouched benchmark. Training ran on Kaggle (2×T4 GPUs) and took 41.2 min for all folds plus a final five-person refit, which has no independent score.

For *known-boundary* results, the window predictions are averaged over each labelled event, given its type, and an event with no usable window counts as an error. Macro-F1 is computed for each held-out participant over the labels present for that person, then averaged equally over participants. In *automatic* sequence evaluation, commands are matched one-to-one to true sign steps at a temporal IoU of at least 0.10. We report the true signs that are detected, detected and correctly named, and also given the correct rule-defined response. A *false command* is an unmatched command whose midpoint lies inside a labelled activity step; wrong names on real signs are not counted as false commands.

TABLE II
HELD-OUT-PARTICIPANT RESULTS

Evaluation slice	People	Events	Score (%)
Known boundaries and type; macro-F1, mean ± s.d. over people			
Isolated activities, all blocks	5	628	50.2 ± 8.4
same folds, ExtraTrees features only	5	628	22.1
Isolated signs, all blocks	5	490	33.2 ± 3.0
Isolated activities, standing/main blocks	5	488	53.0 ± 7.3
Isolated signs, standing blocks	5	350	36.5 ± 5.6
Isolated seated activities	2	140	19.4 ± 8.8
Isolated seated signs	2	140	20.0 ± 6.5
Sequence activity steps	5	208	52.4 ± 13.5
Sequence sign steps	5	208	24.7 ± 7.6
Automatic, sequence signs; pooled share of true signs			
Detected	5	121/208	58.2
Detected and correctly named	5	39/208	18.8
Correct rule-defined response	5	32/208	15.4
239 commands; 50 false commands in 1,218 s of routine activity			

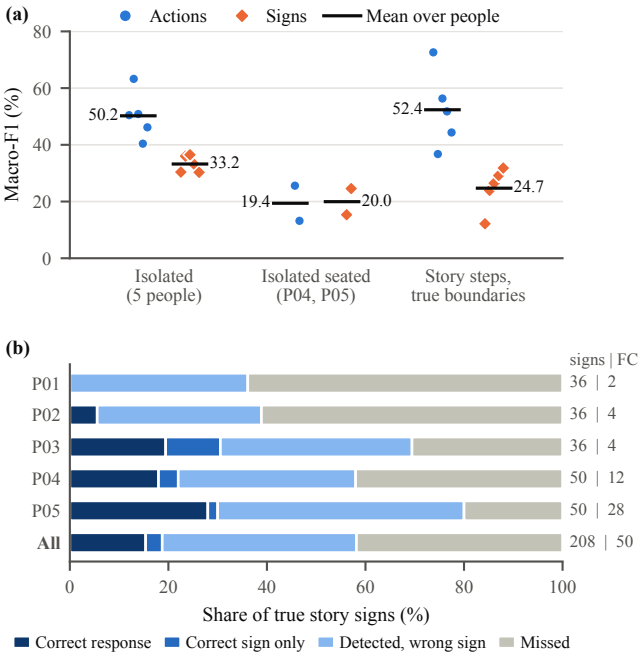


Fig. 3. (a) Macro-F1 per held-out participant when event boundaries and type are supplied. (b) Outcome of every true sequence sign under automatic recognition; FC = false commands during routine activity. P01 has no correctly named sequence sign.

IV. Preliminary Results

Known boundaries. Table II and Fig. 3(a) summarize recognition when the event boundaries are supplied. Pooled over participants, 311 of 628 isolated activities (49.5%) and 171 of 490 isolated signs (34.9%) are named correctly, against 8.3% and 7.1% for uniform guessing. The classes differ widely. Walk is recognized in 46/46 trials, pick up in 40/48 and COOK in 25/35, whereas TV remote (6/67), chopping (4/40), THANK-YOU (1/35), WATER (4/35) and TOILET (4/35) are rarely named correctly. An ExtraTrees classifier on handcrafted

features, refitted on the same training participants of each fold, reaches only 22.1% isolated-activity macro-F1. In the P05 fold, the validation-selected ExtraTrees fusion lowered that participant's activity F1 from 52.6% (CNN alone) to 46.2%; Table II reports the output of the protocol as run.

On sequence steps with true boundaries, activity identity (52.4%) stays close to the isolated level, but sign identity falls to 24.7%. Seated movements are the weakest. For P04 and P05, on the same seven activity classes, 35 of 140 seated trials are named correctly against 66 of 140 standing trials; for signs the counts are 35 of 140 seated against 59 of 140 standing. Posture is confounded with block order in this protocol, so we report this as an observation rather than a measured posture effect.

Automatic recognition. Without supplied boundaries, the system detects 58.2% of the 208 sequence signs but names fewer than a third of the detected signs correctly (Fig. 3(b)). Results vary strongly across participants. P05 has the most correct responses (14/50) but also 28 of the 50 false commands at its selected $\tau = 0.30$, while P01 ($\tau = 0.85$) has no correctly named sign. Extrapolated, the overall false-command count corresponds to about 148 per hour of routine activity, based on only 20.3 min of observation. This count belongs to an operating point chosen to maximize sign-identity F1, which rewards detection more than rejection, so it should not be read as a deployment setting. A higher τ trades detections for fewer false commands; reporting that trade-off on validation participants is part of our ongoing work (Section VI).

V. Error Analysis and Discussion

Sign versus activity is the main loss. With true boundaries, the model's own state decision labels 71 of the 208 sequence signs as activity and 14 as idle. If the type is taken from this decision instead of being supplied, the number of correctly named sequence signs falls from 67 to 38, almost exactly the 39 achieved automatically. Signs in the sequences come straight after activities, are often made seated, and sit in a running stream. The isolated sign blocks also differ from the activity blocks in file and gap labelling, which may make the isolated sign/activity decision easier than it is in the sequences.

Confusions are between similar small motions. TV remote is most often predicted as eat (17/67), chopping as stirring (10/40), WATER as HOT (9/35) and THANK-YOU as HELLO (8/35). These are short hand or arm movements made at similar positions. From the saved predictions alone we cannot tell whether the cause is the signal, the cue-based labels or the model.

Comparison with an earlier version. On the 684 events shared with an earlier pipeline that was evaluated on P01–P03 only, isolated macro-F1 is 54.9% against 48.6% for activities and 34.3% against 20.8% for signs, but sequence activity F1 is lower (55.2% against 60.5%). The earlier pipeline trained on two people per fold and ours on four, and the preprocessing and selection also differ, so this is not an architecture ablation.

Limitations. Five participants, one room and one fixed link rule out any broad generalization claim. P04–P05 were recorded on a different day from P01–P03 and at lower frame rates, so day and participant are confounded. The sequences are scripted, with activities followed by signs. As a result, the anchor activity before a sign always belongs to the same template, so the context rule is tested under favourable ordering, and context accuracy is not measured on its own. The labels come from cues and key presses rather than verified video annotation, and the protocol was revised after an earlier evaluation. The recognizer is offline because its windows are centred, no real device was controlled, and no CSI and wearable data were captured together.

VI. Ongoing Work and Conclusion

Under held-out-participant evaluation on one Wi-Fi link, activities and signs can be named well above chance when their boundaries are known, but finding and naming signs automatically in continuous recordings is still weak. Most of the loss lies in the sign/activity decision, especially for seated and subtle movements. Our next steps, none of which have been run yet, are: reviewing the video and boundaries for the weakest classes; command rejection based on confidence and stability, selected on validation participants and reported as detections, misses, false commands and delay together; a separately reported per-user calibration protocol; and, once the design is frozen, a new session and a new participant for confirmation.

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